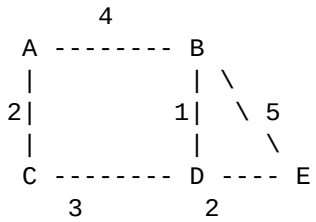
Dijkstra's Algorithm — Step-by-Step Example

Find the shortest distance from vertex A to every other vertex.

Example Graph



Edges: A→B=4, A→C=2, C→D=3, B→D=1, B→E=5, D→E=2.

Step 1 — Initialize

Vertex	Distance from A	Previous
A	0	—
B	∞	—
C	∞	—
D	∞	—
E	∞	—

Starting vertex: A. dist[A]=0 and all other distances are ∞.

Step 2 — Extract-Min

The smallest unvisited distance is A=0. EXTRACT-MIN(Q) → A. This does not mean subtraction; it means select/remove the vertex having the minimum distance. S={A}.

Step 3 — Relax A's neighbors

A→B=4: 0+4=4, so dist[B]=4 and previous[B]=A. A→C=2: 0+2=2, so dist[C]=2 and previous[C]=A.

Step 4 — Extract-Min

Unvisited distances are B=4, C=2, D=∞, E=∞. Minimum is C. EXTRACT-MIN(Q) → C. S={A,C}.

Step 5 — Relax C→D

2+3=5. Since 5<∞, update dist[D]=5 and previous[D]=C.

Step 6 — Extract-Min

Unvisited: B=4, D=5, E=∞. Minimum is B. EXTRACT-MIN(Q) → B. S={A,C,B}.

Step 7 — Relax B's neighbors

B→D=1: 4+1=5, equal to the existing 5, so no improvement. B→E=5: 4+5=9, so update dist[E]=9 and previous[E]=B.

Step 8 — Extract-Min

Unvisited: D=5, E=9. Minimum is D. EXTRACT-MIN(Q) → D. S={A,C,B,D}.

Step 9 — Relax D→E

5+2=7. Since 7<9, update dist[E]=7 and previous[E]=D.

Step 10 — Final Extract-Min

Only E remains. EXTRACT-MIN(Q) → E. All vertices are now processed.

Final Result

Destination	Shortest Distance	Shortest Path
A	0	A
B	4	A → B
C	2	A → C
D	5	A → C → D
E	7	A → C → D → E

Key Concepts

EXTRACT-MIN: Select the unvisited vertex with the smallest current distance.

RELAX: Check whether going through the current vertex gives a shorter path.

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if dist[v] > dist[u] + weight(u,v):
    dist[v] = dist[u] + weight(u,v)
    previous[v] = u
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Remember: Extract-Min = choose the next vertex. Relax = try to improve its neighbors.